



OPEN Sentinel for confidence-aware multi-object tracking

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Multi-object tracking faces two persistent challenges: managing detector confidence and preventing track loss under prolonged occlusions. We introduce Sentinel, an uncertainty-aware tracker that diagnoses per-track uncertainty online and proactively optimizes its tracking strategy. Sentinel consists of two components. Confidence Aware Association (CAA) dynamically reweighs the association-cost terms according to the current track state, enabling the effective use of low-confidence detections while suppressing identity switches. Survival Boosting Mechanism (SBM) preserves tracks at risk of disappearance by exploiting weak detection signals to bridge long occlusions, thereby reducing fragmentation and maintaining identity continuity. Evaluations on MOT17, MOT20, and DanceTrack demonstrate that Sentinel achieves strong performance in Higher Order Tracking Accuracy (HOTA), Identification F1-score (IDF1), and Association Accuracy (AssA), demonstrating its strength in identity preservation and association quality. While this design introduces modest computational overhead and may increase false positives when exploiting low-confidence detections, Sentinel improves robustness in realistic, crowded environments by moving beyond passive reliance on detector outputs to uncertainty-driven, per-track optimization.

Keywords Multi-object tracking, CAA, SBM, State dependent weighted association

Multi-object tracking (MOT) is a computer-vision task for understanding dynamic environments. It detects multiple objects in video sequences, maintains a consistent identity for each object over time, and estimates their trajectories. Its importance continues to grow as a core technology across a wide range of applications, including autonomous driving, intelligent surveillance, sports analytics, and robotics. The tracking-by-detection (TBD)^{1,2} paradigm has become a mainstream approach in modern MOT research. TBD decomposes tracking into per-frame object detection and a temporal association stage that links detection to trajectories. Advances in deep learning have markedly improved this paradigm: high-performance detectors increase the reliability of the detection itself, while refined appearance feature extractors enhance interobject discriminability and thus AssA. As a result, recent MOT methods have achieved state-of-the-art performance on benchmark datasets^{3,4}. Despite these gains, MOT still fails to fully handle the complexity and unpredictability of real-world scenarios. Heavy reliance on detector outputs leads to two entrenched issues that undermine long-term stability and reliability. The first is the detector-confidence dilemma⁵. Conventional TBD pipelines typically restrict their association to high-confidence detections. While this reduces errors from false positives (FPs) and improves precision, it also discards valid low-confidence detections, thereby degrading recall. Under occlusion or motion blur, such low-confidence detections are excluded from the tracklets, inflating false negatives (FNs) and fragments. To address this issue, ByteTrack⁶ proposed a two-stage association strategy that leverages low-confidence detections. Although this substantially reduces FNs and improves tracking recall, it introduces a new challenge: admitting a large volume of low-confidence candidates increases the ambiguity between true objects and background noise, increasing the association complexity and risk of ID-switches (IDSW) driven by FPs. Ultimately, managing detection confidence is not merely a matter of tuning thresholds; it requires a principled approach to interpret and control the uncertainty of each detection. The second problem is the vulnerability of track lifecycle management^{7–9}. Various MOT studies employ linear motion models, such as the Kalman filter, to predict the next position of an object and use a simple age-based management method that terminates the corresponding tracklet when the object has not been observed for a certain number of frames^{10,11}. Age-based approaches may be effective for short occlusions or linear motion; however, they are vulnerable to long-term occlusions, abrupt direction changes, and nonlinear motion, which frequently occur in real-world environments. Consequently, tracklets are unnecessarily split into multiple fragments, and multiple identities are assigned to the same object; consequently, evaluation metrics such as IDF1 and HOTA¹² are degraded.

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We propose the use of Sentinel to address these problems. Sentinel departs from the conventional approach, which passively relies on detection outputs, and presents a paradigm that actively optimizes the tracking strategy around the uncertainty faced by each trajectory. To this end, we introduced two strategies. The first is CAA. CAA uses the confidence score of a detected object not only as a filtering criterion but also as information to regulate uncertainty in the association process. High-confidence detections are reliably linked to existing tracks through a strong association that combines appearance features and motion information, whereas low-confidence detections are used only to recover lost tracks, considering the potential risk of FP. This differential association increases recall by leveraging low-confidence detections while effectively suppressing side effects such as IDSW. The second is a SBM. SBM dynamically manages a survival score by considering each track's state and surrounding context instead of using a fixed frame count. When a tracklet becomes unobserved, SBM updates the survival score by integrating the track's last observed position, velocity, and occlusion likelihood. When the score remains above a certain level, the track remains active, enabling immediate reassociation when the object reappears.

We demonstrated the superiority of the proposed Sentinel through experiments on standard benchmark datasets, such as MOT17¹³, MOT20, and DanceTrack¹⁴. The experimental results show that Sentinel achieved the highest performance on metrics other than Multiple Object Tracking Accuracy (MOTA) in MOT17 and MOT20 and attained remarkable performance on DanceTrack as well. These results indicate that compared with existing state-of-the-art trackers, Sentinel not only exhibits competitive overall tracking accuracy but also delivers consistent improvements in tracking continuity and identity preservation.

Related work

Reliability and uncertainty management in data-association methods

Data association is central to MOT because it assigns the objects detected in each frame to the correct trajectories along the time axis. Early MOT studies sought to solve association problems by using motion models. SORT¹⁵ uses the Kalman filter¹⁶ to predict object positions and adopts Intersection over Union (IoU) between the predicted bounding box and the actual detected bounding box as the sole association criterion. Although SORT's approach is fast and efficient, it has the limitation that it easily misses the ID of the object when the object's motion deviates from the prediction or when occlusion occurs. To overcome these limitations, subsequent studies have begun to exploit appearance information. DeepSORT improves AssA by extracting each object's appearance features with a pretrained Re-ID model and combining them with motion information¹⁷, thereby mitigating the problem of IDSW that occur when visually similar objects are in close proximity. By contrast, many studies, including DeepSORT, have adopted a scheme that combines motion and appearance information with fixed weights^{18,19}. As a result, the same combination ratio is applied to all scenes, revealing a limitation in which association is attempted uniformly without distinguishing between a situation in which an object is tracked stably for a long period and a situation immediately after occlusion in which the position is uncertain. Another axis of data association concerns how to trust and utilize the detector's outputs. Early trackers adopted a conservative strategy of using only high-confidence detections to avoid FPs. This approach increased the precision of tracking but sacrificed recall by discarding detections that were actual objects yet measured with low confidence, and induced track fragments^{20–24}. One study that brought about a change to this paradigm was ByteTrack. ByteTrack proposed a two-stage association strategy that does not discard low-confidence detections but uses them to recover trajectories that are unmatched in the first association. This approach effectively addresses the missed-detection problem and dramatically improves MOT performance. However, ByteTrack's uniform handling causes opposite side effects depending on the state of the trajectory. Stable trajectories induce unnecessary associations with background noise, thereby increasing the risk of identity switching. Unstable trajectories immediately before disappearance may fail to provide a robust association in a timely manner, leading to premature termination. Recent studies have also attempted to address detection and association challenges from broader perspectives. Few-shot object detection via class encoding and multi-target decoding²⁵ addresses detection reliability under limited supervision, and pose-aware association methods such as PB-MOT²⁶ demonstrate that geometric cues can enhance 3D multi-object tracking. However, these approaches focus on improving specific components and do not directly resolve the fundamental issue of adapting association strategies to per-trajectory uncertainty. Therefore, rather than simply adjusting the confidence threshold or relying on auxiliary modalities, an association strategy that dynamically diagnoses the per-trajectory uncertainty and adapts accordingly is required.

Occlusion handling and long-term tracking stability

Whereas data association focuses on frame-to-frame linkages, trajectory lifecycle management focuses on ensuring identity persistence over the entire time axis^{27,28}. Long-term occlusion is the primary factor that undermines identity consistency, and its handling determines a tracker's long-term stability. Lifecycle management commonly used in MOT is age-based termination²⁹. When a tracker fails to find an object corresponding to a given trajectory for a specified number of frames, it completely deletes the trajectory from memory. Although this method is easy to implement, it can be fatal in long-occlusion scenarios. To mitigate this problem, auxiliary techniques, such as Camera Motion Compensation (CMC) and Trajectory Interpolation, have been proposed^{30–32}. Occlusion handling has also been investigated in autonomous multi-agent systems, including ground vehicle tracking in wireless motion capture environments³³ and UAV tracking in low-altitude airspace³⁴. These studies provide complementary perspectives on trajectory survival and identity continuity that may inform pedestrian tracking research. However, these techniques are often confined to specific scenarios or applied uniformly without a fundamental diagnosis of the cause of trajectory loss; thus, they do not provide a general solution. More recently, trackers have partially compensated for the drawbacks of age-based termination by leveraging low-confidence detections to provide opportunities for recovering lost trajectories. Nevertheless,

they still remain passive schemes that wait for matching until a trajectory is terminated, lacking a mechanism that diagnoses the trajectory's risk state in advance and responds proactively.

Proposed method

Existing tracking methodologies have limitations in that they either use fixed association rules based on detection confidence or manage all trajectories using identical termination rules³⁵. We present a paradigm that selectively utilizes detection information centered on the state and uncertainty of each trajectory and actively manages the lifecycle of trajectories by leveraging two components: CAA and SBM.

CAA

Existing trackers calculate data-association costs by combining the appearance and motion information with fixed weights. This static weight approach evaluates all trajectories and situations using the same criteria; thus, it fails to reflect the individual circumstances and uncertainties faced by each trajectory. Nevertheless, when a single association rule is uniformly applied, IDSW frequently occurs when objects with similar appearances are in close proximity, as shown in Fig. 1(a).

To address this issue, we propose CAA. The core of CAA is to diagnose the state of each trajectory in real time and dynamically adjust the weights of the data-association cost function, as shown in Fig. 1(b). CAA classifies the state of trajectories into three categories: Confident, Critical, and Uncertain, based on the information accumulated during the lifecycle of each trajectory, as expressed in Eq. (1).

$$s_i^t = \begin{cases} \text{Confident} & \text{if } hit_i^t \geq \tau_c \wedge \bar{p}_i^t > \theta_c, \\ \text{Critical} & \text{if } miss_i^t \geq \tau_m \vee \bar{p}_i^t < \theta_m, \\ \text{Uncertain} & \text{otherwise} \end{cases} \quad (1)$$

In Eq. (1), state is determined by synthesizing the number of consecutive tracking successes and failures of the corresponding trajectory and the recent confidence score. The Confident state is assigned when both of the following conditions are satisfied: The number of consecutive tracking successes hit_i^t of trajectory i must be equal to or greater than the threshold τ_c , and the average detection confidence $\bar{p}_i^t$ must exceed the threshold θ_c . Trajectories in this state are assumed to have a very high motion prediction accuracy when using the Kalman filter. The Critical state is entered when one of the following conditions is satisfied: This occurs when the number of consecutive tracking failures $miss_i^t$ of trajectory i is equal to or greater than the threshold τ_m or when $\bar{p}_i^t$ is below the threshold θ_m . This indicates a state with a high risk of termination, where the object is in a long-term occlusion state or the detector fails to stably detect the object. An Uncertain state corresponds to all trajectories that do not belong to the Confident or Critical states. It includes newly generated trajectories or trajectories experiencing intermittent tracking failures and is the most common state.

The dynamic weighted association cost according to the trajectory state is expressed in Eq. (2).

$$\begin{aligned} C_{ij} &= \omega_{iou}(s_i^t) d_{ij}^{iou} + \omega_{app}(s_i^t) d_{ij}^{app} + \omega_{ang} d_{ij}^{ang} + \omega_{conf} d_i^{conf} \\ d_{ij}^{iou} &= 1 - IoU(b_i, b_j) \\ d_{ij}^{app} &= 1 - \cos(f_i, f_j) \\ d_{ij}^{ang} &= \frac{1}{4} \sum_{k=1}^4 \frac{\arccos(\text{clip}(\langle v_{i,k}, u_{ij,k} \rangle, -1, 1))}{\pi} \\ d_i^{conf} &= \left| \tilde{p}_i^t - p_i^t \right| \tilde{p}_i^t = 2p_i^{t-1} - p_i^{t-2} \end{aligned} \quad (2)$$

In Eq. (2), the association cost C_{ij} is defined as the weighted sum of the four terms. The IoU-based position mismatch term d_{ij}^{iou} reflects the lack of overlap between the predicted bounding box b_i and candidate b_j as a penalty. The appearance-mismatch term d_{ij}^{app} represents the appearance difference based on the cosine similarity between the Re-ID embeddings f_i and f_j . The movement direction consistency term d_{ij}^{ang} evaluates the motion direction consistency by normalizing the angle between the recent trajectory movement direction



Fig. 1. Process of resolving IDSW through dynamic association strategy of confidence-aware association.

and the vector connecting the predicted position and the actual observed position in the current frame. $v_{i,k}$ is the estimated movement direction vector in the recent k -th interval of trajectory i , $u_{i,j,k}$ is the vector pointing to the position of the current frame prediction candidate j , and clip is an operation that restricts the inner product value to $[-1, 1]$. The confidence fluctuation term d_i^{conf} measures the difference between the linearly extrapolated confidence $\bar{p}_i^t$ and current confidence p_i^t , where p_i^t is the detection confidence associated with trajectory i at frame t . Because this term is independent of the candidate j , it is applied uniformly to all candidate associations for trajectory i , effectively modulating the overall association willingness based on the temporal stability of detections. This regularization prevents the dynamic weighting mechanism from switching rapidly between association modes due to transient detection failures, thereby maintaining temporally stable state transitions. The d^{iou} weight ω^{iou} and the d^{app} weight ω^{app} are applied variably according to s_i^t , while the weights ω^{ang} and ω^{conf} for d^{ang} and d^{conf} are set as fixed constants independent of state since they are auxiliary terms for search constraint and stabilization. The Confident state increases the weight of the IoU distance and decreases the weight of the appearance distance, because the reliability of the position prediction is high. This prevents IDSW by ensuring that, even if other objects with similar appearances appear nearby, candidates that deviate significantly from the predicted position are not associated. The Critical state reduces the reliability of the position information because the prediction error of the Kalman filter likely accumulates when it is not observed for a long period. Instead, because the validity of the last observed appearance features is maintained, the weight of the IoU distance is decreased and the weight of the appearance distance is increased to search a wider area and increase the probability of detecting objects with similar appearances. The Uncertain state performs standard association by considering the position and appearance information in a balanced manner. State-dependent weight adjustment directly reflects the uncertainty of each trajectory in the association process, thereby prioritizing ID preservation in the Confident state and expanding reconnection opportunities in the Critical state to enhance overall tracking performance.

SBM

Trajectory lifecycle management in MOT universally relies on age-based methods that terminate trajectories that have not been observed for a specific period. This passive approach reveals the fundamental vulnerabilities in long-term occlusion situations³⁶. When an object is occluded by other objects or crowds and reappears, it commits the critical error of prematurely terminating the corresponding trajectory and assigning a new ID to the reappearing object. This causes unnecessary track fragmentation, as shown in Fig. 2 (a), and is a major cause of the degradation of key metrics such as IDF1. To overcome these limitations, we propose an SBM that actively rescues trajectories at risk of termination. SBM is an active hypothesis verification mechanism that departs from the existing paradigm of simply managing trajectories by survival time. Instead, it establishes survival hypotheses for each trajectory and statistically verifies them using low-confidence detection information. In other words, SBM establishes the hypothesis that a lost trajectory still exists within the scene and extends the life of the trajectory by discovering the most likely candidate from the low-confidence detection candidate pool.

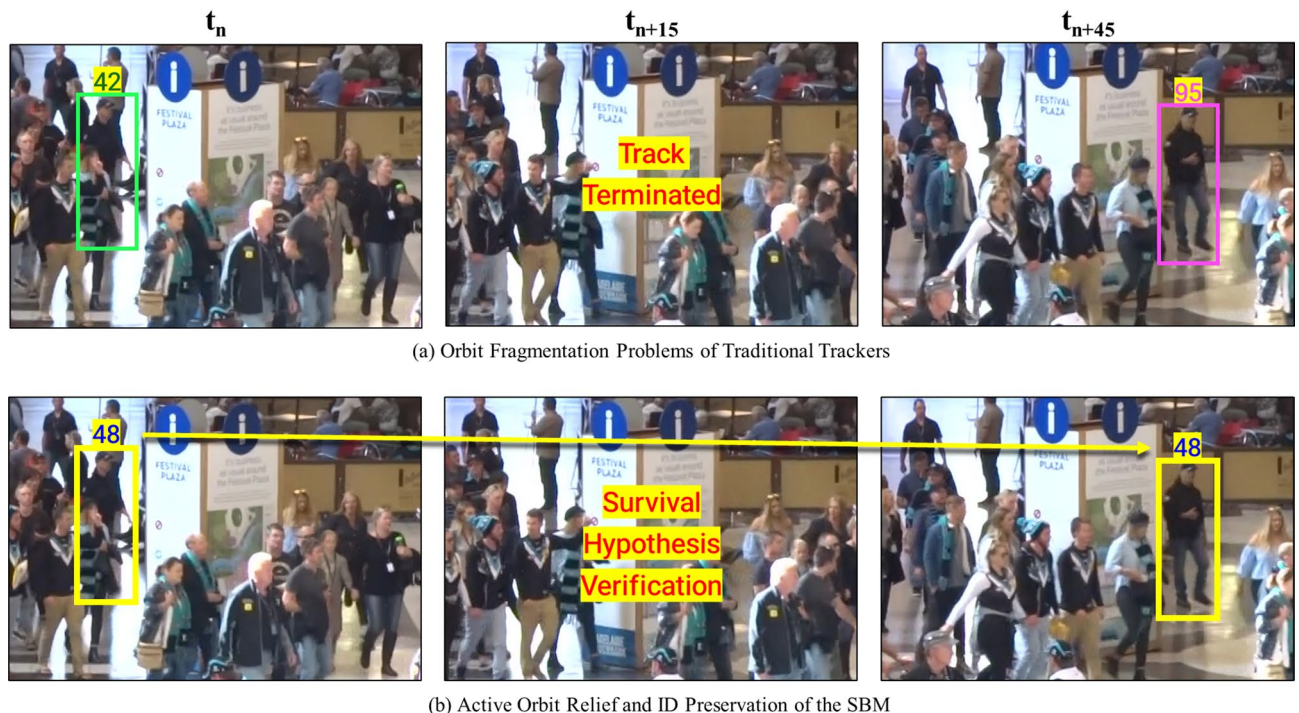


Fig. 2. Response to long-term occlusion and securing trajectory continuity through survival boosting mechanism.

The first step in SBM is to quantify the uncertainty and termination risks of the lost trajectories. Therefore, we introduce a state variable D that explicitly models the reconnection necessity of the trajectories, as shown in Eq. (3).

$$D_i^t = \max \left(D_{\min}, \min \left(\frac{\text{miss}_i^t}{\tau_D}, 1 \right) \right) \quad (3)$$

miss_i^t represents the number of frames in which trajectory i has consecutively failed to associate up to frame t , and τ_D denotes the reference number of frames at which D reaches its maximum value. The lower bound of D is set to $D_{\min} (> 0)$ and the upper bound to 1. When a trajectory fails to associate and miss_i^t accumulates, D_i^t increases linearly, and when successfully connected with a candidate object, miss_i^t is reset to 0 and D_i^t decreases. Because this value is re-evaluated for every frame, it can respond sensitively to dynamic situational changes such as abrupt scene changes, crowd congestion, and occlusion. $D_{\min} > 0$ serves as a safety margin. Linear prediction models such as the Kalman filter are effective in stationary or constant-velocity motion sections; however, it is easy to overtrust the prediction in sections where nonlinear movements such as sudden stops or sharp turns occur³⁷. If $D = 0$ were allowed, it would indicate absolute trust in the prediction model, which would weaken the alternative hypothesis search and pose a high risk that accumulated detection failures would eventually lead to ID fragmentation. Therefore, the positive lower bound maintains a minimum alert state in the tracker, continuing the alternative hypothesis search even in stable tracking sections, thereby improving resilience in situations where the prediction fails. After diagnosing the risk level of trajectories through D , SBM defines the possibility that a low-confidence detection candidate j is the same object as the lost trajectory i as a survival hypothesis and verifies it by quantifying it as a survival score S_{ij} . S_{ij} is calculated as the weighted sum of spatial similarity IoU_b , appearance similarity $\cos(f_i, f_j)$, and kinematic constraint Π_{ij} , and dynamically adjusts the weight ω of each term according to D as shown in Eq. (4). Specifically, as D_i^t increases, indicating prolonged occlusion, the emphasis shifts from position-based cues to appearance-based cues, reflecting the increased positional uncertainty.

$$S_{ij} = \omega_{iou} (D_i^t) \text{IoU}_b(i, j) + \omega_{app} (D_i^t) \cos(f_i, f_j) + \omega_{mov} (D_i^t) \Pi_{ij} \quad (4)$$

The first component, IoU_b , is expressed in Eq. (5).

$$\text{IoU}_b(i, j) = \text{IoU} \left(\text{expand} \left(\hat{b}_i^t, \beta_i^t \right), b_j \right) \quad (5)$$

$$\beta_i^t = \beta_0 + \kappa D_i^t$$

IoU_b is a spatial similarity metric for compensating for the prediction uncertainty of the Kalman filter that occurs in long-term occlusion situations. The standard IoU is effective when the object movement is linear and the occlusion is short; however, when a trajectory is not observed for a long time, the state covariance of the Kalman filter increases, and the position accuracy of the predicted bounding box $\hat{b}_i^t$ deteriorates. In this state, excluding potential reconnection candidates solely because the overlap between the predicted position and the actual detection region b_j is small becomes a major cause of tracking failure. To solve this problem, IoU_b forms a flexible search gate that accommodates the prediction uncertainty by calculating IoU after expanding both the predicted and detection bounding boxes by the same ratio. The expansion ratio β_i^t increases linearly in proportion to D_i^t . This approximation models the possibility that the object deviates further from the predicted position as the loss period increases. β_0 is the basic expansion rate that compensates for the minimum prediction error that can occur even in short-term occlusion situations, and κ is a hyperparameter that controls the rate at which uncertainty increases over time. Consequently, IoU_b plays the role of maximizing the research success rate of lost trajectories by adaptively adjusting the search area according to the state of the trajectory.

The second component, $\cos(f_i, f_j)$, was measured as the cosine similarity between the appearance feature embeddings extracted using the Re-ID model. f_i represents the appearance feature vector when trajectory i was last observed and f_j is the feature vector of low-confidence detection candidate j . ByteTrack applies low-confidence detections in the same manner to all unmatched trajectories that were missed in the first matching stage, thereby increasing the risk of false associations with background noise, even for stable trajectories. In contrast, SBM is activated only for trajectories in the Critical state and utilizes appearance similarity as a core cue for survival hypothesis verification. This state-dependent weight adjustment expands the reassociation opportunities while suppressing false associations compared with ByteTrack's static association method. Particularly, in conditions where multiple objects with similar appearances exist, appearance information becomes a decisive cue that resolves association ambiguity, which is difficult to address using spatial cues alone. Only candidates for which S_{ij} exceeds the predefined threshold θ pass the first validity verification, and by verifying the survival hypothesis based on the last observed high-confidence appearance information, confusion with incorrect low-confidence background noise or other objects is prevented.

The third component, Π_{ij} , is as shown in Eq. (6). Π_{ij} operates as a constraint that evaluates the relationship between the predicted position of track i and the observed position of candidate j to filter out associations that are physically impossible or have remarkably low plausibility in advance. Even if the appearance is similar, if an object has moved a distance that is physically difficult to reach within a single frame, it is highly likely to have an incorrect association. Π_{ij} determines whether it is within a realistic motion range based on the Euclidean distance between the observed center point c_j and the predicted center point $\hat{c}_i$, and can include normalization tailored to scene-specific motion limits such as previous velocity or the maximum speed of pedestrians. At this

time, Π_{ij} is a scalar gate with a range of $[0,1]$, where a value of 0 means candidate j is physically excluded for track i , and a value greater than 0 maintains it as an association candidate. Furthermore, Π_{ij} is directly reflected as the motion plausibility term of S_{ij} and is combined with spatial proximity and appearance similarity through a state-dependent weighted sum.

$$\begin{aligned}\Pi_{ij} &= \max(0, 1 - \frac{\|c_j - \hat{c}_i\|}{d_{max}}) \\ c_j &= \left[\frac{x_{1j} + x_{2j}}{2}, \frac{y_{1j} + y_{2j}}{2} \right]^T \\ \hat{c}_i &= [\hat{x}_i, \hat{y}_i]^T\end{aligned}\quad (6)$$

The identified low-confidence candidate j^* is not matched immediately but undergoes a selective candidate promotion procedure to prevent damage to global optimality owing to greedy decisions. First, the first validity is verified by confirming whether the calculated highest survival score S_{ij^*} exceeds θ , thereby blocking indiscriminate promotion of implausible candidates. The confidence scores of the candidates that passed verification were adjusted using Eq. (7).

$$p_{j^*}^+ = \min(1, p_{j^*} + \mu_1 + \mu_2 D_i^t) \quad (7)$$

μ_1 is a static confidence correction term, and $\mu_2 D_i^t$ operates as a dynamic correction term proportional to the trajectory's disappearance period. Thus, the disappearance risk is lowered with priority while adjusting the trade-off with the FP risk within a limited range. Only when the elevated score $p_{j^*}^+$ satisfies the high-confidence criterion is candidate j^* incorporated into the high-confidence candidate group and subsequently competes for final matching with equal qualification to other high-confidence detections in the state-dependent association stage of CAA.

Pipeline

Sentinel operates according to the procedure shown in Fig. 3 and Algorithm 1: When the current frame is provided, the CMC is applied to reduce the prediction errors caused by global motion. The trajectories from the previous frame predict the current object state using the Kalman filter, and the detection results are separated into high-confidence and low-confidence sets according to the confidence criteria. The state of each trajectory is classified as Confident, Critical, or Uncertain using the CAA. The classification process is performed by comprehensively analyzing the number of consecutive tracking successes, the number of consecutive tracking failures, and the recent average confidence. The results are utilized as criteria for adjusting the weights of association costs and determining whether to activate SBM. At this time, SBM is applied to trajectories classified as Critical. SBM searches for candidates associated with Critical trajectories within the low-confidence detection set, selects valid candidates, adjusts the confidence scores upward, and then includes them in the final detection candidate group. By contrast, trajectories in the Confident and Uncertain states proceed to the next association stage without going through SBM. In the CAA-Weighted Association stage, an association is performed between the final detection candidate group and all trajectories whose states have been classified. The association cost is calculated using a cost function that is dynamically weighted according to the CAA state of each trajectory. Confident trajectories prioritize position information, whereas Critical trajectories prioritize appearance information. When the association is complete, the matched trajectories update the Kalman filter state, unassociated detections are initialized as new trajectories, and long-term unobserved trajectories are removed according to the lifetime policy.

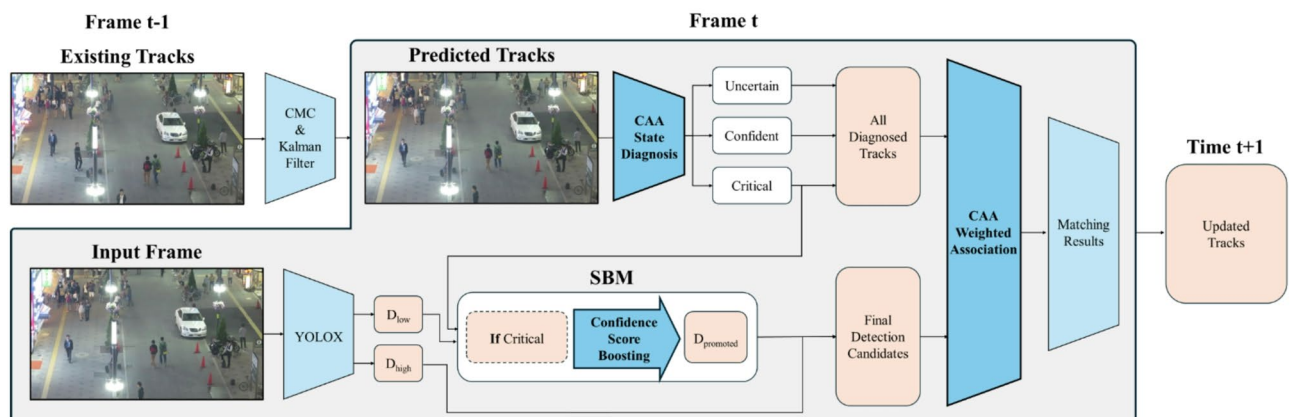


Fig. 3. Sentinel pipeline overview.

Input:

Video sequence V ; Object detector DET ; Kalman filter KF ;
Confidence thresholds τ_h, τ_l ; Track initialization threshold ε ;
CAA parameters ϕ_{caa} ; SBM parameters ϕ_{sbm}

Output:

Tracks T of the video

```

1: Initialization:  $T \leftarrow \phi$ 
2: for each frame  $f_k \in V$  do
3:    $D_k \leftarrow DET(f_k)$ ;  $D_h \leftarrow \phi$ ;  $D_l \leftarrow \phi$ 
4:
5:   ApplyCMC( $T$ )
6:   for  $t$  in  $T$  do  $t$ 
7:      $KF.predict(t)$ 
8:   end for
9:
10:  DiagnoseCAASates( $T, \phi_{caa}$ )
11:
12:   $T_{crit} \leftarrow \{t \in T \mid t.state == Critical\}$ 
13:   $D_{promoted} \leftarrow ActivateSBM(T_{crit}, D_l, \phi_{sbm})$ 
14:
15:   $D_{finalcandidates} \leftarrow D_h \cup D_{promoted}$ 
16:   $M, U_{trk}, U_{det} \leftarrow AssociateCAA(T, D_{finalcandidates}, \phi_{caa})$ 
17:  UpdateTracks( $T, M, U_{trk}$ )
18:
19:  for  $d$  in  $(U_{det} \cap D_h)$  do
20:    if  $d.conf \geq \varepsilon$  then  $T \leftarrow T \cup InitTrack(d)$  end if
21:  end for
22:
23:   $T \leftarrow RemoveObsolete(T)$ 
24: end for
return  $T$ 

```

Algorithm 1. Sentinel tracking pipeline.

Experimental and results

Datasets

We conducted experiments on three standard benchmark datasets—MOT17, MOT20, and DanceTrack—to validate the performance of Sentinel. MOT17 is a standard benchmark consisting of seven training and seven test sequences that include diverse lighting conditions, camera movements, and pedestrian densities, and it plays a key role in evaluating the overall robustness and generalization performance of algorithms. MOT20 focuses on crowded pedestrian environments and provides scenarios in which severe interobject occlusions frequently occur. Such environments present extreme conditions in which trackers are prone to cause IDSW. DanceTrack poses various challenges to existing pedestrian datasets. It is characterized by dancers with similar appearances exhibiting nonlinear and unpredictable movements, which invalidates the assumptions of linear motion models such as the Kalman filter and creates an environment where appearance similarity is maximized.

Metric

We used standard evaluation metrics in the MOT field to comprehensively evaluate the performance of Sentinel. As primary metrics, we adopted MOTA, IDF1, and HOTA to quantitatively analyze the various aspects of tracking performance. MOTA is a traditional metric that measures tracking accuracy based on FP, FN, and IDSW. It is highly dependent on detection performance and is useful for identifying the basic error level of a tracker. However, it has the limitation of placing greater weight on detection accuracy rather than on association quality. IDF1 focuses on evaluating ID preservation performance. It measures how long each trajectory consistently maintains the correct ID, thereby demonstrating the effectiveness of the proposed CAA and SBM in preventing trajectory fragmentation and suppressing IDSW in long-term occlusions. HOTA has recently been recognized as the most balanced evaluation metric. It explicitly separates and calculates detection and association accuracy and derives the final score through the geometric mean of the two. This allowed for an independent yet comprehensive evaluation of the quality of detection and association, showing the highest correlation with human visual assessment.

Implementation details

For object detection, we adopted YOLOX-X³⁸, which is widely used in MOT research. The detector was pretrained on the COCO dataset and the hyperparameters were optimized for the training dataset of each benchmark. For appearance feature extraction, we used SBS50³⁹ from the FastRe-ID framework. The appearance feature extractor was pretrained on each dataset, and the default settings of the framework were used. During the tracking process, we utilized both the standard Kalman filter and the CMC to improve the prediction accuracy of the linear motion model. The cost matrix for data association was calculated using the weighted summation of four components: IoU, cosine, confidence, and angular distance, and was dynamically adjusted. To find the

Hardware Specifications		Software Versions	
CPU	Intel Core i9-10900 K	Operating system	Ubuntu Linux 20.04.3 LTS (https://ubuntu.com)
Graphics card	NVIDIA GeForce RTX 3090	Programming language	Python 3.9.7 (https://www.python.org)
SSD	Samsung SSD 980 PRO 1 TB	GPGPU	CUDA 11.4.r11.4 (https://developer.nvidia.com/cuda-toolkit)
RAM	Samsung DDR5-4800 64GB	Deep neural network library	cuDNN 8.6.0 (https://developer.nvidia.com/cudnn)
		Deep learning framework	PyTorch 1.12.1 (https://pytorch.org)

Table 1. Hardware specifications and software versions.

Method	MOTA (%)	IDF1 (%)	HOTA (%)	MT (%)	Rcll (%)	IDSW
EnhanceCenter ⁴⁰	70.4	66.3	55.1	38.9	76.4	-
BUSCA ⁴¹	78.6	79.2	63.9	48.5	83.2	1,428
MAA ⁴²	79.4	75.9	62.0	57.6	86.2	1,452
RTU + + ⁴³	79.5	79.1	63.9	55.3	85.0	1,302
StrongSORT++	79.6	79.5	64.4	53.6	84.7	1,194
RAP-SORT ⁴⁴	80.0	79.4	64.4	-	-	-
ConfTrack ⁴⁵	80.0	81.2	65.4	58.7	87.7	1,155
REAL-SORT ⁴⁶	80.1	79.1	64.7	-	-	-
BoTSORT	80.5	80.2	65.0	54.4	84.8	1,212
ByteTrack	80.6	78.9	63.6	57.6	87.0	1,239
UCMCTrack	80.6	81.0	65.7	58.9	87.3	1,689
SparseTrack ⁴⁷	81.0	80.1	65.1	54.6	85.5	1,170
StrongTBD	81.6	80.8	65.6	55.7	86.0	954
PIA2 ⁴⁸	82.2	81.1	66.0	57.3	87.6	1,026
Sentinel	80.3	82.0	66.0	61.1	88.3	921

Table 2. Performance comparison of Sentinel against state-of-the-art trackers on MOT17 test set.

optimal matching pairs, we used a strategy of sequentially connecting clear pairs through an iterative allocation method that starts with an initial threshold and then decreases it. The experimental setup is listed in Table 1.

Experimental results

Table 2 shows that Sentinel achieved overall superior performance on the MOT17 test dataset. It recorded 66.0 in HOTA, achieving the highest score alongside PIA2, and a competitive score of 80.3 in MOTA, confirming its excellent ability to manage basic tracking errors. Notably, it achieved the best performance for IDF1, Mostly Tracked (MT), and Recall (Rcll), which are directly related to ID preservation and long-term stability, and recorded the lowest number of IDSW. The highest Rcll is the result of effectively utilizing low-confidence detection, which differs from ByteTrack’s approach of uniformly associating all low-confidence candidates. ByteTrack attempts to associate all candidates without a state distinction, which leads to numerous IDSW owing to associations with background noise or incorrect objects. In contrast, SBM selectively utilizes low-confidence detection only for Critical-state trajectories immediately before loss to recover the trajectories. This approach, which minimizes side effects from unnecessary association attempts and reduces FNs, reduces IDSW by approximately 25% and improves IDF1 by 3.1% compared with ByteTrack. Furthermore, CAA’s dynamic association strategy played a crucial role in minimizing IDSW. Trackers such as StrongSORT and BOTSORT use a static association method that combines motion and appearance information with fixed weights. Although this method is effective in stable tracking situations, it has limitations in situations with high positional prediction uncertainty, such as immediately after occlusion. CAA diagnoses the trajectory state in real time, increasing the weight of motion information in the Confident state to conservatively suppress IDSW and increasing the weight of appearance information in the Critical state to enable a broader search. This ability to dynamically optimize the association strategy according to the situation enables a more robust handling of various environmental changes compared to trackers that rely on fixed rules, leading to the lowest IDSW. These improvements in IDSW, MT, and IDF1 collectively indicate suppression of track fragmentation. In contrast to ByteTrack, which accumulates fragmentation errors due to uniform low-confidence associations, Sentinel’s selective candidate promotion through SBM prevents unnecessary trajectory splits while CAA’s state-dependent weighting maintains stable identity assignment.

Table 3 presents the experimental results for the MOT20 dataset. Sentinel achieved the best performance with a HOTA of 65.6, an IDF1 of 81.0, and recorded the lowest number of IDSW at 920. In addition, MT and Rcll exhibited the best performance. This proves that individual identities are maintained most stably, even in extreme situations where objects overlap and are difficult to identify⁵⁴. The excellent performance of Sentinel is attributed to overcoming the limitations in trajectory lifecycle management shown by existing trackers in extreme crowding and long-term occlusion situations, which are the core challenges of MOT20. Most trackers

Method	MOTA (%)	IDF1 (%)	HOTA (%)	MT (%)	Rcll (%)	IDSW
EnhanceCenter	47.1	49.3	38.5	16.6	-	2,339
BUSCA	72.7	76.3	61.8	59.7	75.6	1,006
OCSORT	75.7	76.3	62.4	65.5	79.5	942
REAL-SORT	76.6	76.3	62.9	-	-	-
RAP-SORT	76.8	79.0	64.2	-	-	-
EscapeTrack ⁴⁹	77.4	80.5	64.8	70.9	83.5	1,061
ByteTrack	77.8	75.2	61.3	69.2	83.1	1,223
BoTSORT	77.8	77.5	63.3	70.2	82.8	1,313
GMMotion ⁵⁰	78.0	77.3	63.2	69.9	83.2	1,212
SparseTrack	78.1	77.6	63.5	70.0	83.3	1,120
MHO ⁵¹	78.2	76.9	62.5	70.4	83.4	956
SuppTrack ⁵²	78.2	75.5	61.9	70.9	84.3	1,325
PIA2	78.5	79.0	64.7	69.2	83.5	1,023
ImprAsso ⁵³	78.6	78.8	64.6	70.8	84.0	992
Sentinel	77.3	81.0	65.6	70.9	84.3	920

Table 3. Performance comparison of Sentinel against state-of-the-art trackers on MOT20 test set.

Method	MOTA (%)	IDF1 (%)	HOTA (%)	AssA (%)
GTR ⁵²	84.7	48.0	50.3	31.9
ByteTrack	89.9	47.7	53.9	32.1
FineTrack ⁵³	89.9	52.7	59.8	38.5
FCG ⁵⁶	89.9	48.7	46.5	29.9
StrongSORT++	91.1	55.6	55.2	38.6
MotionTrack	91.3	58.2	58.6	41.7
GHOST	91.3	56.7	57.7	39.8
CBIoU ⁵⁷	91.6	60.6	61.6	45.4
BASE ⁵⁸	91.7	56.4	-	-
OCSORT	92.0	55.1	54.6	38.3
CMTrack ⁵⁹	92.5	61.8	63.3	46.4
Sentinel	93.4	62.8	65.7	47.5

Table 4. Performance comparison on DanceTrack test set.

use age-based termination methods that cause trajectory fragmentation when objects are occluded by crowds for several seconds or longer. For example, ImprAsso and PIA2 focus on optimizing the association matching process and securing association robustness through the aggregation of past information, respectively. However, their trajectory lifecycle itself relies on age-based management, making them fundamentally vulnerable to long-term occlusion. By contrast, SBM, instead of passively waiting for termination when a trajectory disappears, establishes a survival hypothesis and actively verifies survival through low-confidence detections. This is an active rescue mechanism that probes the survival possibility of trajectories without relying on fixed time thresholds and is a factor that prevents errors of prematurely terminating trajectories owing to long-term occlusion and suppressed trajectory fragmentation, which other trackers commit. This is also different from the reactive approach of OCSORT, which retrospectively corrects the state of the reconnected trajectories after occlusion. Whereas OCSORT focuses on recovering already broken trajectories, SBM takes a preventive approach that utilizes low-confidence detection to prevent situations where trajectories break first, thereby securing high ID continuity⁵⁵. The highest HOTA was attributed to the synergistic effect of SBM and CAA. When SBM robustly maintains trajectory life in long-term occlusion situations, CAA dynamically lowers the weight of motion information for objects in Uncertain states emerging from occlusion and relies on appearance information to stably reconnect them. This combination achieved high accuracy in both detection and association aspects and is why Sentinel is significantly ahead in core indicators of tracking association quality, such as HOTA, IDF1, and IDSW, despite ImprAsso recording the highest MOTA score. The combination of the lowest IDSW and highest MT further demonstrates that Sentinel suppresses track fragmentation even under extreme crowding, validating the preventive design of SBM over conventional reactive approaches.

Table 4 lists the experimental results for the DanceTrack test dataset. Sentinel achieved the best performance, recording the highest performance in all metrics and surpassing all the compared trackers. This performance superiority stems from the adaptive association strategy of Sentinel, which responds to situations in which the reliability of information fluctuates dynamically. Trackers such as MotionTrack and OCSORT rely heavily on Kalman-filter-based linear motion prediction. However, in the face of sudden directional changes or vigorous

dance movements, these linear assumptions fail and directly lead to association errors. CBIOU uses a heuristic that expands the matching space by setting a buffer to compensate for this; however, it fails to resolve the fundamental modeling limitations. In contrast, CAA performs association by assigning a higher weight to appearance information when tracking failures of a trajectory accumulate and the reliability of the motion information is judged to be low. This ability to diagnose the trajectory state in real time and dynamically optimize the information utilization strategy is the key mechanism that overcomes the limitations that other trackers, which inherently rely on fixed rules, cannot overcome. In addition, models that focus on sophisticated appearance features, such as FineTrack and GHOST, encounter limitations in discriminability when dancers wearing similar costumes form groups or overlap. In these situations, Sentinel has a dual defense mechanism. First, when a dancer shows a stable trajectory for a moment, CAA conversely increases the weight of the motion information to prevent the ID from being transferred to other dancers with similar appearances nearby. Second, when frequent and long occlusions occur as dancers form groups, SBM actively rescues trajectories in danger of termination through hypothesis verification using low-confidence detection, preventing unnecessary trajectory fragmentation⁶⁰. It is noteworthy that Sentinel achieved the highest MOTA score of 93.4, even though all the comparison trackers, except GTR, used the same detector. The highest MOTA was a remarkable achievement, considering that it recorded a relatively low MOTA compared with the other trackers in MOT17 and MOT20. This superiority originates from the interaction between DanceTrack's unique characteristics and Sentinel. The unpredictable movements and frequent occlusions of DanceTrack are the main causes of existing trackers prematurely terminating trajectories and accumulating numerous FN errors. By contrast, SBM reduces the FN that frequently occurs in other trackers by actively utilizing low-confidence detection information to recover trajectories in danger of loss. In other words, the reduction in FN made a decisive contribution to MOTA improvement. Simultaneously, CAA suppresses IDSW, thus reducing other major error items in the MOTA. In conclusion, the high MOTA proved that Sentinel achieved additional performance in reducing tracking-specific errors while responding to specific challenges in the dataset.

Table 5 presents the stepwise application effects of CAA and SBM. After applying CAA to each benchmark, HOTA commonly increased, which is interpreted as a result of the state-awareness-based association strategy that structurally reduces false connections in crowded sections and improves association quality. Subsequently, when SBM was added, IDF1 consistently increased, which was explained by the suppression of disconnections in the reconnection stage after non-observation, thereby strengthening ID preservation. In MOT17, even amid the diversity of scene conditions, sequential improvement is observed, where CAA reduces association ambiguity in general scenes to stabilize the HOTA baseline and SBM-reinforced ID preservation by alleviating residual disconnections in intermittent occlusion and reappearance sections. In the MOT20, where the frequency of crowding and long-term occlusion is high, reconnection stability determines success or failure. CAA suppresses false connections between nearby objects to solidify the association quality, and SBM is effective in reducing fragmentation by alleviating critical situations occurring at the boundary before and after trajectory loss. In DanceTrack, nonlinear motion and high appearance similarity coexist, resulting in frequent sections where the reliability of linear motion-based prediction is low. Under these conditions, CAA preemptively improved HOTA by refining ambiguous associations through state-dependent weight adjustment, and IDF1 also increased concurrently. Subsequently, when SBM was combined, the disconnections in the reappearance section after occlusion decreased, leading to an additional increase in IDF1, and HOTA improved slightly. To evaluate real-time applicability, FPS measurements are reported following standard MOT evaluation protocols. Only tracking components excluding detection time were measured, and the average FPS over three independent runs after a 10-frame warm-up is presented. Applying CAA reduces FPS from 126.1 to 100.5 on MOT17, from 23.8 to 17.7 on MOT20, and from 622.3 to 588.9 on DanceTrack. This reduction is attributed to real-time state diagnosis that classifies each trajectory into Confident, Critical, or Uncertain states, as well as dynamic weight computation for association costs. The relatively smaller reduction on DanceTrack compared to pedestrian-dense benchmarks is attributed to fewer trajectories per frame, which reduces the cumulative cost of per-trajectory state diagnosis. When SBM is added, noticeable additional reductions are observed across all datasets. MOT17 decreases from 100.5 to 81.5 FPS, MOT20 from 17.7 to 14.1 FPS, and DanceTrack from 588.9 to 539.9 FPS. This overhead arises from the survival hypothesis verification process, which requires computing spatial similarity with expanded bounding boxes, appearance similarity through Re-ID embeddings, and kinematic constraints for candidate

Dataset	CAA	SBM	IDF1 (%)	HOTA (%)	FPS
MOT17-val			86.63	69.90	126.1
	✓		87.13	70.44	100.5
	✓	✓	88.29	70.87	81.5
MOT20-val			88.34	68.20	23.8
	✓		89.49	68.78	17.7
	✓	✓	89.78	69.11	14.1
DanceTrack-val			65.20	62.12	622.3
	✓		67.45	63.64	588.9
	✓	✓	67.76	63.89	539.9

Table 5. Ablation study on validation sets of MOT17, MOT20, and DanceTrack to analyze the contribution of each component of Sentinel.

	Hyper parameters		HOTA(%)	AssA(%)	IDF1 (%)		Hyper parameters		HOTA(%)	AssA(%)	IDF1 (%)
CAA	τ_c	40	63.82	50.55	67.61	SBM	τ_D	1	63.84	50.57	67.71
		50	63.34	49.78	67.06			5	63.85	50.59	67.72
		45	63.89	50.65	67.76			10	63.89	50.65	67.76
	θ_c	0.65	63.47	50.09	67.42		β_0	35	63.70	50.34	67.52
		0.70	63.89	50.65	67.76			40	63.89	50.65	67.76
		0.70	63.09	49.25	66.58		θ	0.90	62.58	48.67	66.19
	θ_m	0.80	62.94	49.13	66.32			1.00	63.64	50.27	67.45
		0.75	63.89	50.65	67.76			0.95	63.89	50.65	67.76

Table 6. Sensitivity analysis of CAA and SBM hyperparameters on the DanceTrack validation set.

	Hyper parameters		HOTA(%)	AssA(%)	IDF1 (%)		Hyper parameters		HOTA(%)	AssA(%)	IDF1 (%)
CAA	τ_c	3	70.85	74.60	88.26	SBM	τ_D	15	69.82	72.77	86.61
		4	70.86	74.61	88.26			25	70.78	74.66	87.99
		2	70.87	74.62	88.29			20	70.87	74.62	88.29
	θ_c	0.75	70.85	74.57	88.19		β_0	40	70.86	74.62	88.28
		0.80	70.87	74.62	88.29			45	70.87	74.62	88.29
		0.60	70.85	74.59	88.21		θ	0.70	70.25	73.65	87.22
	θ_m	0.70	70.86	74.62	88.28			0.80	70.79	74.21	87.68
		0.65	70.87	74.62	88.29			0.75	70.87	74.62	88.29

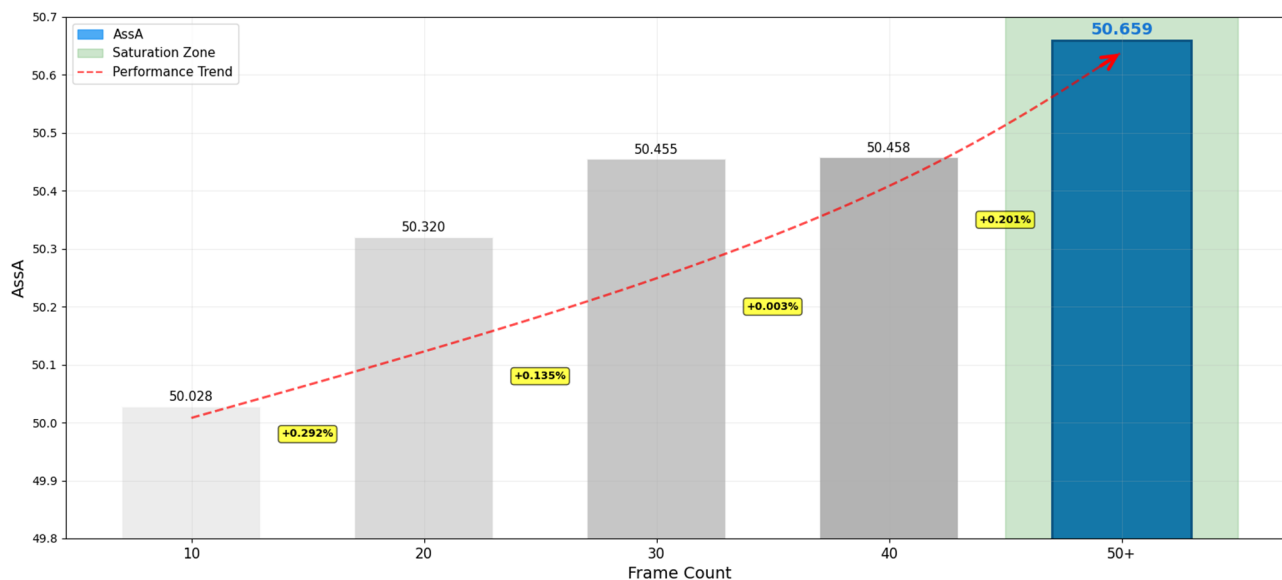
Table 7. Sensitivity analysis of CAA and SBM hyperparameters on the MOT17 validation set.

evaluation. Although SBM is primarily activated for Critical-state trajectories, candidate search within the low-confidence detection pool and survival score computation introduce measurable costs across all datasets. The relatively smaller reduction on DanceTrack is attributed to fewer objects per frame compared to MOT17 and MOT20, and the characteristics of dance scenarios where occlusions are resolved within relatively short durations, resulting in a lower proportion of trajectories reaching the Critical state. Despite these computational costs, Sentinel maintains real-time performance suitable for practical applications. Specifically, 81.5 FPS on MOT17 and 539.9 FPS on DanceTrack comfortably exceed real-time requirements, while MOT20's 14.1 FPS remains reasonable given its extreme crowd density where even baseline tracking operates at only 23.8 FPS. In summary, CAA serves as a foundational element that systematically elevates association accuracy through state-aware cost weighting, and SBM acts as a complementary element that reinforces ID preservation in long-term occlusion and re-entry sections. These two components work synergistically to produce consistent increases in HOTA and IDF1 across all three datasets, with the performance gains justifying the computational overhead in scenarios where identity preservation is critical.

Table 6 presents the sensitivity analysis of CAA and SBM hyperparameters on the DanceTrack validation set. For CAA, three parameters governing state classification were examined. τ_c achieved optimal performance at 45. A value of 40 caused premature Confident state transitions before reliable motion predictions were established, whereas 50 delayed the transition unnecessarily. θ_c showed optimal performance at 0.70, whereas 0.65 classified low-confidence tracks as Confident, causing inappropriate weight application. θ_m achieved optimal performance at 0.75. A value of 0.7 triggered Critical transitions even with minor confidence drops, causing unnecessarily aggressive matching, whereas 0.8 delayed Critical transitions, resulting in late responses to problematic tracks. For SBM, τ_D showed stable performance across values of 1, 5, and 10, with marginal improvement at 10. β_0 achieved optimal results at 40, whereas 35 provided insufficient expansion ratio to compensate for baseline prediction errors caused by nonlinear motion. θ performed best at 0.95. A value of 0.90 increased incorrect matches due to the low survival score threshold, resulting in AssA dropping to 48.67, whereas 1.00 missed legitimate reconnection opportunities through overly strict verification, resulting in AssA dropping to 50.27.

Table 7 presents the sensitivity analysis results on the MOT17 validation set. For CAA, τ_c achieved optimal performance at a lower value of 2, as the predictable linear motion patterns of pedestrians allow Confident state transitions with shorter tracking histories. θ_c performed best at 0.80, higher than 0.75, indicating that higher appearance distinctiveness among pedestrians allows stricter criteria. θ_m showed optimal results at 0.65, where 0.6 triggered Critical transitions too frequently and 0.7 delayed the transitions. For SBM, τ_D achieved optimal performance at 20, higher than DanceTrack, as the linear motion of pedestrians maintains valid position predictions longer during occlusion, making gradual increase of D appropriate. β_0 performed best at 45, indicating that a wider baseline search region is required in crowded pedestrian environments. θ achieved optimal performance at 0.75, lower than DanceTrack's 0.95, indicating that higher appearance diversity among pedestrians allows permissive verification to expand reconnection opportunities. The difference in optimal values across datasets indicates that Sentinel can adaptively respond to diverse environmental characteristics;

SBM	FP	FN
Off	15,696	32,630
On	16,968	30,000

Table 8. FP-FN trade-off analysis of SBM on MOT20 validation set.**Fig. 4.** Performance analysis of AssA on the DanceTrack validation set according to variations in non-observation tolerance frames.

DanceTrack's nonlinear motion and high appearance similarity require conservative state transitions and strict candidate verification, whereas MOT17's predictable pedestrian dynamics permit faster state classification and more permissive reconnection criteria.

Table 8 presents the FP and FN changes when SBM is activated on the MOT20 validation set. With SBM enabled, FP increases from 15,696 to 16,968, representing an 8.1% increase, whereas FN decreases from 32,630 to 30,000, representing an 8.1% reduction. Notably, the absolute FN reduction of 2,630 is approximately 2.07 times greater than the absolute FP increase of 1,272, indicating that SBM recovers more true objects than it incorrectly admits. This trade-off stems from the design principles of SBM. SBM explores low-confidence detections only for trajectories in the Critical state and restricts candidate acceptance through survival score threshold θ and selective promotion procedures. This conservative verification mechanism suppresses the FP increase to 8.1% while effectively reducing FN caused by long-term occlusion. In the extreme crowding environment of MOT20, detection misses are a primary cause of trajectory fragmentation. Consequently, FN reduction has a substantial impact on overall tracking quality. The improvements in IDF1 and HOTA observed in Table 5 when SBM is applied demonstrate that this FN reduction translates into enhanced ID preservation and association quality.

Figure 4 shows the change in AssA according to the change in the number of consecutive tracking failure tolerance frames, which is a hyperparameter of SBM. The X-axis represents the number of non-observation frames allowed until trajectory termination, and the y-axis represents AssA. As the tolerance increases, AssA increases gently and converges to approximately 50.66 in the range above 50 frames. This rising and converging tendency is attributed to the following factors. The cause of the increase in AssA was analyzed as the effect of three mechanisms. First, the survival window is expanded. Increasing tolerance allows trajectories to be maintained longer, increasing opportunities for survival hypothesis verification, and strengthening reconnection attempts. Second, the search area is adaptively expanded. As tolerance frame count increases, D_{it} increases linearly and expands the prediction box at the same ratio, absorbing the positional uncertainty caused by long-term occlusion and reducing the omission of potential reconnection opportunities. Third, a false-positive suppression mechanism exists. Candidates must pass, and even after passing, confidence is conservatively adjusted to compete fairly well with other candidates in the final association stage. This process effectively controls false connections with background noise caused by indiscriminate search expansion. Convergence above a specific frame is interpreted as an upper-limit effect formed by the data distribution and design constraints. Owing to the tendency for long-term occlusion duration to be resolved within approximately 2.5 s in DanceTrack, when tolerance is expanded to that level, the majority that can be recovered are already reconnected. In the subsequent range, the proportion of scene exit cases and trajectories with large appearance and motion drift increases, making the reconnection probability low, even when an extended search is applied. Internal constraints, such as clipping for safety, also gradually reduce marginal gains; therefore, the additional tolerance above a specific

frame is offset on average, flattening the curve. Consequently, as the marginal utility of the tolerance increment is exhausted, AssA approaches its practical upper limit.

Figure 5 demonstrates the improvement of the ID-related metrics using the state-dependent weight adjustment method compared with the static method. This performance improvement was attributed to the following two mechanisms. First, decision-making reflects the asymmetry of false connection costs. During the tracking process, incorrect high-confidence connections serially distort the cost matrix of subsequent frames, causing ID theft and serial IDSW, leading to near-irreversible losses. In contrast, temporary FN occurring from withholding associations with low-confidence candidates have fewer serial ripple effects, resulting in relatively small losses. The conservative candidate promotion of SBM linked with CAA exploits this asymmetry. That is, instead of immediately connecting candidates lacking confidence, by granting only fair competition qualifications to the high-confidence candidate group, it accepts the small cost of a short-term FN increase and maximizes the large benefit of long-term ID preservation. Second, matching stabilization is achieved by resolving joint ambiguity. In crowded environments, the reliability of both appearance and motion information deteriorates simultaneously, generating numerous similar-cost candidates within the cost matrix. CAA diagnoses the trajectory state in real time, dynamically redistributes the weights of each cost term, and expands the cost gap between the top and second-best candidates. This increases the decision margin of the Hungarian assignment, structurally suppressing the possibility of rank reversal due to minute noise or uncertainty, thereby reducing IDSW. In conclusion, the two mechanisms operate complementarily, functioning as a principle by which CAA stably maintains ID continuity, even in crowded situations.

Figure 6 demonstrates the performance improvement of SBM through a comparison with the existing two-stage matching. SBM significantly reduces FP and IDSW and improves HOTA and MT, demonstrating superiority in long-term tracking stability. This superior performance originates from its robust ability to cope with long-term occlusions and nonlinear motion. Existing two-stage matching attempts to uniformly associate low-confidence candidates, absorb background noise, or terminate trajectories with long-term occlusion owing to fixed termination rules, accumulating false positives. In contrast, SBM responds with a single strategy that expands the search area according to the situation while performing conservative verification; as the loss period lengthens, it progressively widens the search radius to secure connection pathways with objects reappearing after a long time. Simultaneously, the matching process after confidence boosting selectively passes only high-confidence candidates, thereby blocking the influx of background noise. As a result, reconnection failures decrease and false connections are suppressed, leading to concurrent increases in HOTA and MT. The effect of SBM is maximized when combined with CAA's state-dependent weight adjustment. Such adaptive cue fusion enables trajectories restored by SBM to be stably reconnected, even in Uncertain states, immediately after occlusion. In summary, the excellent performance of SBM originates from the synergy of search space expansion prepared for long-term occlusion, false connection suppression through conservative verification, and real-time weight redistribution by CAA. The organic combination of the two mechanisms structurally reduces FP and IDSW, resulting in improvements in HOTA and MT.

Figure 7 presents a qualitative comparison between the baseline tracker and Sentinel on DanceTrack, where nonlinear motion and high appearance similarity make ID preservation differences most visually apparent. In the baseline tracker, the target object is initially assigned ID 2 at Frame 32. However, when the object enters an occlusion state at Frame 94, the baseline tracker completely loses the trajectory. When the object reappears at Frame 101, the tracker fails to recognize it as the same object and assigns a new ID 12, causing track fragmentation. This phenomenon demonstrates the fundamental vulnerability of age-based termination

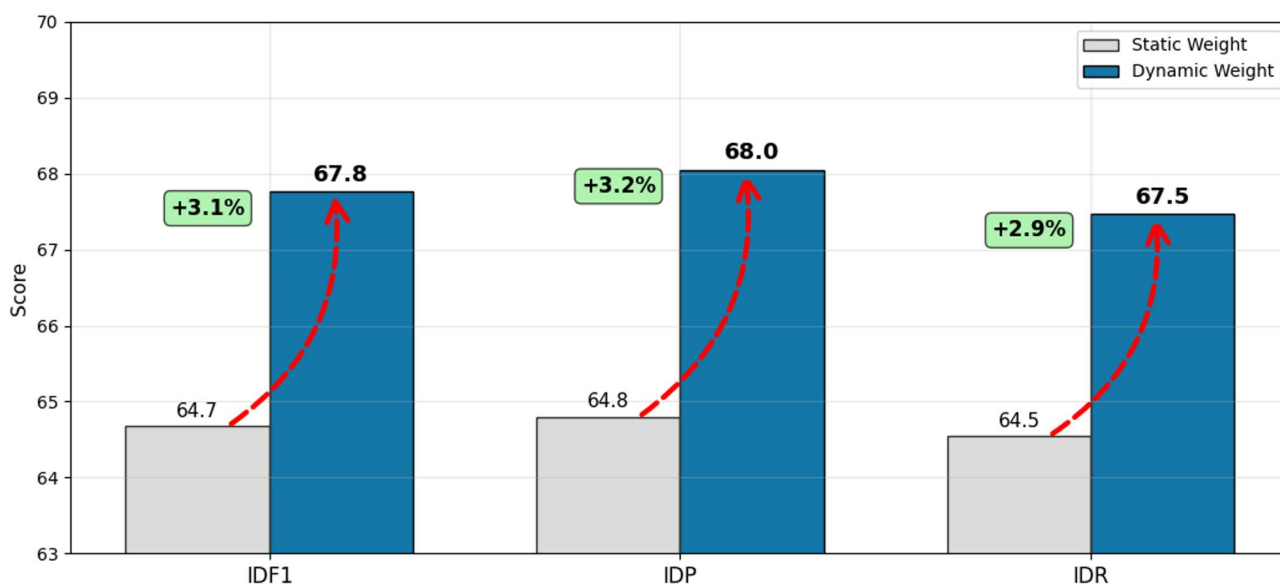


Fig. 5. Comparison of ID-related metrics on the DanceTrack validation set between static method and state-dependent weight adjustment method.

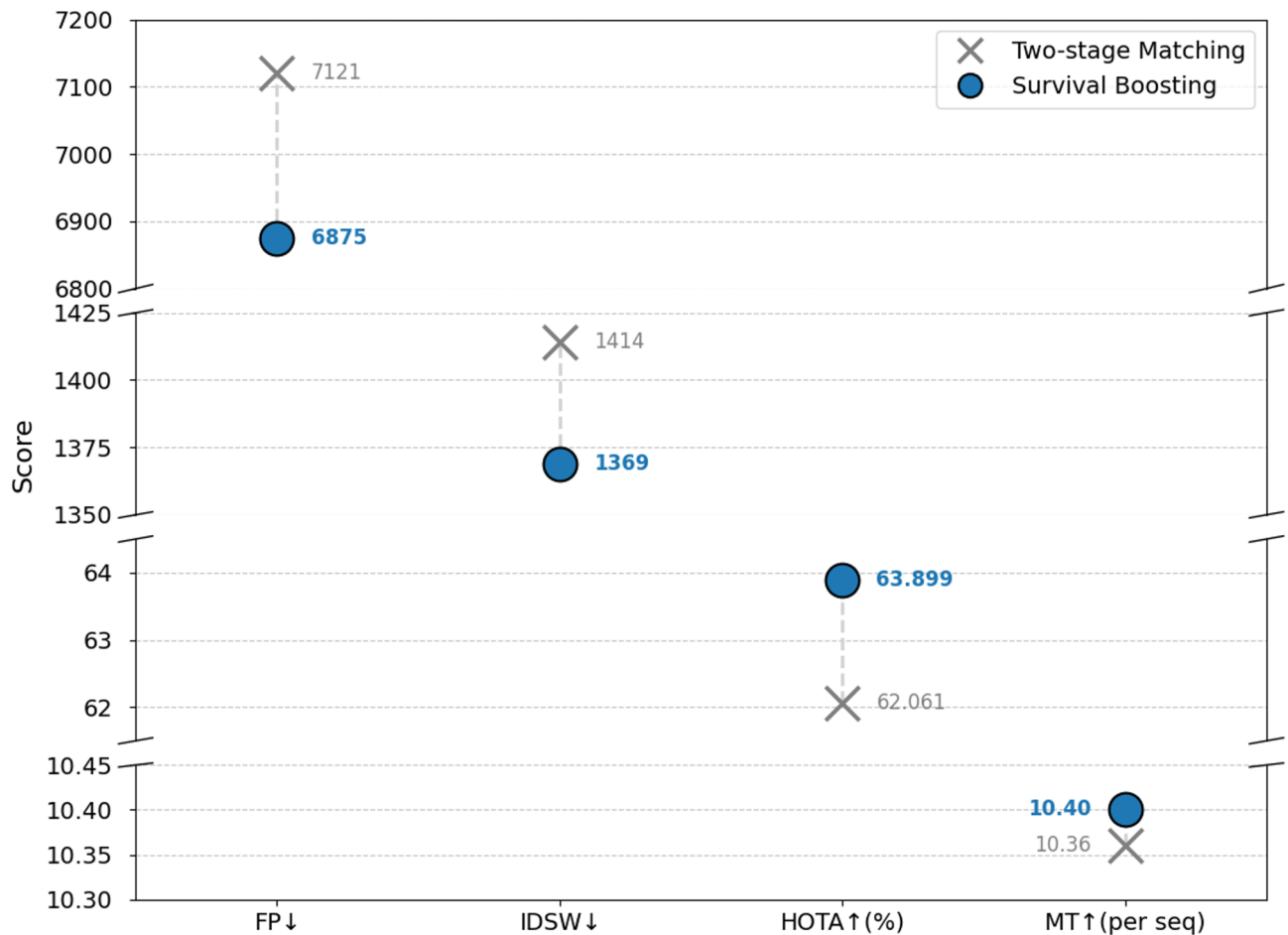


Fig. 6. Comparison of long-term tracking stability on the DanceTrack validation set between SBM and existing two-stage association method.

methods that prematurely terminate trajectories under long-term occlusion. In contrast, Sentinel maintains a consistent ID 2 throughout the entire sequence. When the object enters an occlusion state, SBM establishes a survival hypothesis and actively searches for reconnection candidates within the low-confidence detection pool. When the object reappears, CAA dynamically increases the weight of appearance information for trajectories in the Critical state, enabling stable reconnection with the original ID. These qualitative results demonstrate that the synergistic operation of CAA and SBM effectively prevents unnecessary ID switches and preserves trajectory continuity even under challenging occlusion scenarios.

Conclusions

We proposed Sentinel, which dynamically optimizes tracking strategies centered on trajectory uncertainty to address the chronic limitations of MOT, namely, the detection confidence dilemma and trajectory extinction vulnerability. Sentinel overcomes the limitations of existing tracking approaches by using two core components: CAA and SBM. CAA diagnoses the state of each trajectory in real time and dynamically adjusts the weights of the association costs, thereby conservatively protecting stable trajectories and actively reconnecting trajectories at risk of loss to minimize IDSW. SBM departs from conventional passive age-based extinction rules and actively validates the survival hypothesis of lost trajectories through low-confidence detections, robustly maintaining trajectory continuity even in long-term occlusion situations. Through extensive experiments on the MOT17, MOT20, and DanceTrack datasets, Sentinel achieved superior performance in HOTA and IDF1, demonstrating its strength in identity preservation and association quality. While Sentinel shows slightly lower MOTA compared to some recent trackers such as PIA2 and StrongTBD, this trade-off aligns with its design philosophy of prioritizing long-term tracking continuity. However, the proposed approach has several limitations. CAA incurs increased computational costs due to real-time state diagnosis and dynamic weight calculation, and the optimal state classification thresholds vary depending on dataset characteristics, requiring hyperparameter re-tuning for new environments. SBM may increase FP due to its aggressive utilization of low-confidence detections, and its effectiveness can be limited in environments where the detector itself has low recall, as fewer candidates are available for survival hypothesis verification. Despite these limitations, Sentinel showed consistent performance superiority in IDF1, which indicates ID continuity, and HOTA, a comprehensive performance metric, in challenging scenarios with extreme crowding, long-term occlusions, and frequent irregular movements. In

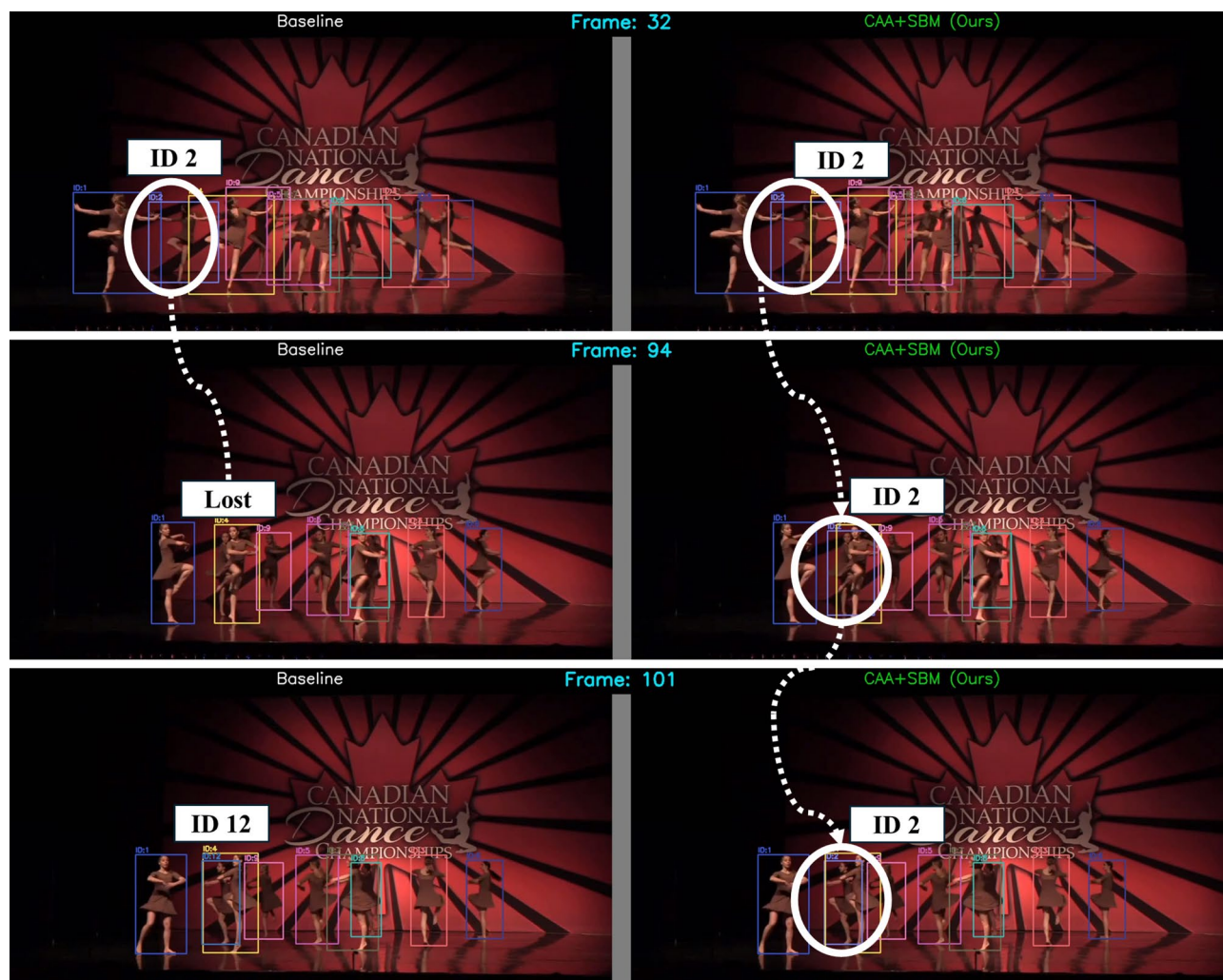


Fig. 7. Qualitative comparison of ID preservation under occlusion between baseline tracker and Sentinel on DanceTrack.

conclusion, Sentinel presents a new paradigm that intelligently adjusts association and lifecycle management strategies by focusing on the intrinsic state of individual trajectories, elevating tracking robustness and accuracy in various real-world environments to the next level.

Data availability

The data supporting the findings of this study are publicly available. The datasets used in this study include MOT17 and MOT20 from the MOT Challenge repository, and the DanceTrack dataset. They can be accessed at the following URLs respectively: <https://motchallenge.net/data/MOT17/>, <https://motchallenge.net/data/MOT20/>, and <https://github.com/DanceTrack/DanceTrack>. The MOT Challenge datasets are provided under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 license (<https://creativecommons.org/licenses/by-nc-sa/3.0/>), while the DanceTrack dataset is available for non-commercial research purposes. All usage of the data in this study complies with the terms of their respective licenses.

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References

- Fengwei, Y. & et al. Poi: Multiple object tracking with high performance detection and appearance feature. *In European conference on computer vision*. 36–42, (2016). https://doi.org/10.1007/978-3-319-48881-3_3
- Bochinski, E., Eiselein, V. & Sikora, T. High-speed tracking-by-detection without using image information. *In 14th IEEE international conference on advanced video and signal based surveillance*. 1–6, (2017). <https://doi.org/10.1109/AVSS.2017.8078516> (2017).
- Yunhao, D. Strongsort: Make deepsort great again. *IEEE Trans. Multimedia*. 25, 8725–8737. <https://doi.org/10.1109/TMM.2023.240881> (2023).
- Nguyen, P., Quach, K. G., Kitani, K. & Luu, K. Type-to-Track: Retrieve Any Object via Prompt-based Tracking. *Proceedings of the 37th International Conference on Neural Information Processing Systems*. 3205–3219. (2024).

5. Liu, Y., Li, Y., Xu, D., Yang, Q. & Tao, W. Adaptive Kalman Filter with power transformation for online multi-object tracking. *Multimedia Syst.* **29** (3), 1231–1244. <https://doi.org/10.1007/s00530-023-01052-7> (2023).
6. Zhang, Y. & et al. Bytetrack: Multi-object tracking by associating every detection box. *European conference on computer vision.* 1–21. (2022). https://doi.org/10.1007/978-3-031-20047-2_1
7. Pang, Z. & et al. SimpleTrack: Understanding and Rethinking 3D Multi-object Tracking. *European Conference on Computer Vision.* 680–696, (2023). https://doi.org/10.1007/978-3-031-25056-9_43
8. Chiu, H. et al. Probabilistic 3D Multi-Modal, Multi-Object Tracking for Autonomous Driving. In *IEEE International Conference on Robotics and Automation.* 14227–14233, 14227–14233, (2021). <https://doi.org/10.1109/ICRA48506.2021.9561754> (2021).
9. Stadler, D. & et al. Improving Multiple Pedestrian Tracking by Track Management and Occlusion Handling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition.* 10958–10967, (2021). <https://doi.org/10.1109/CVPR46437.2021.01081>
10. Aharon, N. & et al. BoT-SORT: Robust Associations Multi-Pedestrian Tracking. *arXiv preprint arXiv:220614651* (2022).
11. Miah, M., Bilodeau, G. A., and Saunier, N. Learning data association for multi-object tracking using only coordinates. *Pattern Recognition* **160**, (2025).
12. Luiten, J. et al. Hota: A higher order metric for evaluating multi-object tracking. *Int. J. Comput. Vis.* **129**, 548–578. <https://doi.org/10.1007/s11263-020-01375-2> (2021).
13. Milan, A., Leal-Taixé, L., Reid, I., Roth, S. & Schindler, K. MOT16: A benchmark for multi-object tracking. *arXiv preprint arXiv:1603.00831* (2016).
14. Sun, P. et al. Multi-object tracking in uniform appearance and diverse motion, In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition.* 20993–21002. (2022). <https://doi.org/10.1109/CVPR52688.2022.02032>
15. Bewley, A., Ge, Z., Ott, L., Ramos, F. & Upcroft, B. Simple online and realtime tracking. In *IEEE international conference on image processing.* 3464–3468, 3464–3468, (2016). <https://doi.org/10.1109/ICIP.2016.7533003> (2016).
16. Kalman, R. E. A new approach to linear filtering and prediction problems. *J. Basic Eng.* **82**(1), 35–45. <https://doi.org/10.1115/1.3662552> (1960).
17. Wojke, N., Bewley, A. & Paulus, D. Simple online and realtime tracking with a deep association metric. *IEEE Int. Conf. Image Process.* <https://doi.org/10.1109/ICIP.2017.8296962> (2017).
18. Liu, H. et al. Improved DeepSORT algorithm based on multi-feature fusion. *Appl. Syst. Innov.* <https://doi.org/10.3390/asi5030055> (2022).
19. Wang, Z., Zheng, L., Liu, Y., Li, Y. & Wang, S. Towards real-time multi-object tracking. In *European conference on computer vision.* 107–122, (2020). https://doi.org/10.1007/978-3-030-58621-8_7
20. Zhang, Y., Wang, C., Wang, X., Zeng, W. & Liu, W. FairMOT: On the fairness of detection and re-identification in multiple object tracking. *Int. J. Comput. Vis.* <https://doi.org/10.1007/s11263-021-01513-4> (2021).
21. Zhou, X., Koltun, V. & Krähenbühl, P. Tracking objects as points. *Eur. Conf. Comput. Vis.* **474** (490). https://doi.org/10.1007/978-3-030-58548-8_28 (2020).
22. Pang, J. & et al. Quasi-dense similarity learning for multiple object tracking. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition.* 164–173, (2021). <https://doi.org/10.1109/CVPR46437.2021.00023>
23. Wu, J. & et al. Track to detect and segment: An online multi-object tracker. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition.* 12352–12361, (2021). <https://doi.org/10.1109/CVPR46437.2021.01217>
24. Liu, Z. et al. Multi-object tracking by performing scene decomposition based on pseudo-depth. *arXiv preprint arXiv:2306.05238* (2023).
25. Guo, X., Yang, H., Wei, M. & Zhang, Y. Few-shot object detection via class encoding and multi-target decoding. *IET Cyber-Systems Rob.* **5** (2), 218–226. <https://doi.org/10.1049/csy2.12088> (2023).
26. Pang, B., Xu, Y., Chen, J. & Li, L. P. B. M. O. T. Pose-aware Association Boosted Online 3D Multi-Object Tracking. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems.* 14170–14177, 14170–14177, (2025). <https://doi.org/10.1109/IROS60139.2025.11247375> (2025).
27. Wang, G., Wang, Y., Zhang, H., Gu, R. & Hwang, J. N. Exploit the Connectivity: Multi-Object Tracking with TrackletNet. *arXiv preprint arXiv:1811.07258* (2018).
28. Pang, B. et al. Adopting Tubes to Track Multi-Object in a One-Step Training Model. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition.* 6307–6317, (2020). <https://doi.org/10.1109/CVPR42600.2020.00634>
29. Kefu, Y. I. et al. UCMCTrack: Multi-Object Tracking with Uniform Camera Motion Compensation. *Proc. AAAI Conf. Artif. Intell.* **38**(7), 6702–6710. <https://doi.org/10.1609/aaai.v38i7.28493> (2024).
30. Guo, X., Zheng, Y. & Wang, D. PMTrack: Multi-object Tracking with Motion-Aware. *Asian Conf. Comput. Vis.* **3091–3106** https://doi.org/10.1007/978-981-96-0960-4_26 (2024).
31. Sinishaw, M. L. & Liu, S. JDECMC: Improving JDE-based multi-object tracking with camera motion compensation. *Displays* <https://doi.org/10.1016/j.displa.2024.102682> (2024).
32. Zeng, K. Noise-control multi-object tracking. *Complex. Intell. Syst.* **9**(4), 4331–4347. <https://doi.org/10.1007/s40747-022-00946-9> (2023).
33. Memon, S. A. et al. Tracking multiple autonomous ground vehicles using motion capture system operating in a wireless network. *IEEE Access* **12**, 60592–60604. <https://doi.org/10.1109/ACCESS.2024.3394536> (2024).
34. Memon, S. A. et al. Tracking multiple unmanned aerial vehicles through occlusion in low-altitude airspace. *Drones* **7**(4), 241. <https://doi.org/10.3390/drones7040241> (2023).
35. Seidenschwarz, J., Braso, G., Elezi, I. & Leal-Taixe, L. Simple Cues Lead to a Strong Multi-Object Tracker. *Conf. Comput. Vis. pattern Recognit.* **13813–13823** <https://doi.org/10.1109/CVPR52729.2023.01327> (2023).
36. Wang, Y. H. & et al. Similarity learning for occlusion-aware multiple object tracking. In *Proceedings of the AAAI Conference on Artificial Intelligence.* 38(6), 5740–5748, (2024). <https://doi.org/10.1609/aaai.v38i6.28386>
37. Xiao, C. et al. Motiontrack: Learning motion predictor for multiple object tracking. *Neural. Netw.* <https://doi.org/10.1016/j.neunet.2024.106539> (2024).
38. Ge, Z. et al. & Yolox: Exceeding yolo series in 2021. *arXiv preprint arXiv:2107.08430* (2021).
39. He, L. et al. Fastreid: a pytorch toolbox for real-world person re-identification. *arXiv preprint arXiv:02631* (2020). (2020). (2006).
40. Yang, H. S., Park, S. W., Jung, S. H. & Sim, C. B. EnhanceCenter for improving point based tracking and rich feature representation. *Sci. Rep.* <https://doi.org/10.1038/s41598-025-88924-2> (2025).
41. Vaquero, L., Xu, Y., Alameda-Pineda, X., Brea, V. M. & Mucientes, M. Lost and found: Overcoming detector failures in online multi-object tracking. In *European Conference on Computer Vision.* 448–466, (2024). <https://doi.org/10.48550/arXiv.2407.10151>
42. Stadler, D. & Beyerer, J. Modelling ambiguous assignments for multi-person tracking in crowds. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision.* 133–142, (2022). <https://doi.org/10.1109/WACVW54805.2022.00019>
43. Wang, S. et al. Extendable multiple nodes recurrent tracking framework with RTU++. *IEEE Trans. Image Process.* **31**, 5257–5271. <https://doi.org/10.1109/TIP.2022.3192706> (2022).
44. Zhang, S., Zhu, Y., Sun, Y., Liu, W. & Huang, Z. RAP-SORT: Advanced multi-object tracking for complex scenarios. *Displays* <https://doi.org/10.1016/j.displa.2026.103361> (2026).
45. Zhang, X., Zhao, H., He, S. & Li, Y. REAL-SORT: Relation-aware for real-time multiple object tracking. *Knowl. Based Syst.* <https://doi.org/10.1016/j.knsys.2026.115373> (2026).

46. Jung, H., Kang, S., Kim, T., Kim, H. & ConfTrack Kalman Filter-Based Multi-Person Tracking by Utilizing Confidence Score of Detection Box. In *Proceedings of the IEEE/CVF Winter Conference on Computer Vision and Pattern Recognition*. (2024). <https://doi.org/10.1109/WACV57701.2024.00645>
47. Liu, Z., Wang, X., Wang, C., Liu, W. & Bai, X. Sparsetrack: Multi-object tracking by performing scene decomposition based on pseudo-depth. *IEEE Trans. Circuits Syst. Video Technol.* <https://doi.org/10.1109/TCSVT.2024.3524670> (2025).
48. Stadler, D. & Beyerer, J. Past information aggregation for multi-person tracking. In *2023 IEEE International Conference on Image Processing*. 321–325, (2023)., October <https://doi.org/10.1109/ICIP49359.2023.10223159> (2023).
49. Yi, K., Wu, H., Hao, W. & Hu, R. EscapeTrack: Multi-object tracking with estimated camera parameters. *Signal. Process.* <https://doi.org/10.1016/j.sigpro.2025.109958> (2025).
50. Chen, B., Wei, Z., Lei, W., Wang, C. & GMMotion Neighborhood Information Matters for Online Multi-Pedestrian Tracking. In *PRICAI 2024: Trends in Artificial Intelligence*. (2024). https://doi.org/10.1007/978-981-96-0122-6_8
51. Gao, Y., Xu, H., Li, J., Wang, N. & Gao, X. Multi-Scene Generalized Trajectory Global Graph Solver with Composite Nodes for Multiple Object Tracking. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(3), 1842–1850, (2024). <https://doi.org/10.1609/aaai.v38i3.27953>
52. Zhang, Y. et al. Handling Heavy Occlusion in Dense Crowd Tracking by Focusing on the Heads. In *AI 2023: Advances in Artificial Intelligence*. 79–90, (2023). https://doi.org/10.1007/978-981-99-8388-9_7
53. Stadler, D. & Beyerer, J. An Improved Association Pipeline for Multi-Person Tracking. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. 3170–3179, (2023). <https://doi.org/10.1109/CVPRW59228.2023.00319>
54. Sun, P. et al. & Global Tracking Transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 8761–8770 (2022).
55. Ren, H. et al. Focus on Details: Online Multi-Object Tracking with Diverse Fine-Grained Representation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 11289–11298, (2023). <https://doi.org/10.1109/CVPR52729.2023.01086>
56. Fang, Y. M. O. T. & FCG++ Enhanced Representation of Spatio-temporal Motion and Appearance Features. *arXiv preprint arXiv:2411.10028* (2024).
57. Qin, Z. et al. Motiontrack: Learning robust short-term and long-term motions for multi-object tracking. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 17939–17948, (2023). <https://doi.org/10.1109/CVPR52729.2023.01720>
58. Yang, F., Odashima, S., Masui, S. & Jiang, S. Hard to Track Objects with Irregular Motions and Similar Appearances? Make It Easier by Buffering the Matching Space. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 4799–4808. (2023). <https://doi.org/10.1109/WACV56688.2023.00478>
59. Larsen, M., Rolfsjord, S., Gusland, D., Ahlberg, J. & Mathiassen, K. BASE: Probably a Better Approach to Visual Multi-Object Tracking. In *Proceedings of the 19th International Joint Conference on Computer Vision*. 110–121 (2024).
60. Shim, K., Hwang, J., Ko, K. & Kim, C. A confidence-aware matching strategy for generalized multi-object tracking. In *IEEE international conference on image processing*. 4042–4048., 4042–4048, (2024). <https://doi.org/10.1109/ICIP51287.2024.10647729> (2024).

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Author contributions

H.S.Y. conceived Sentinel and designed CAA and SBM. S.W.P. and C.B.S. implemented the tracking pipeline and conducted experiments on the MOT17, MOT20, and DanceTrack datasets. S.H.J. supervised the project and contributed to manuscript revision. All authors have reviewed and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Informed Consent

This study utilizes three publicly available datasets: MOT17 and MOT20 from the MOT Challenge benchmark, and the DanceTrack dataset. The MOT Challenge datasets consist of fully anonymized data, for which no additional informed consent is required. The DanceTrack dataset was collected with prior informed consent from all participants. All datasets were used in accordance with their terms and conditions, which permit their use for research purposes and in academic publications.

Additional information

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